

PUBLIC HEALTHCARE DATA, BUILT FOR DECISIONS

Provider Intelligence Data Platform

A production-grade foundation that transforms fragmented federal datasets into trustworthy provider, practice, market, industry, and clinical-research intelligence.

16	7.37M	22.44M	594,772
registered sources	core providers	provider-drug records	clinical studies

Validated production release snapshot - July 22, 2026. Every change requires explicit approval, bounded smoke validation, and retention of a complete rollback release.

01 Platform at a glance

The Provider Intelligence Data Platform is the canonical public-data plane for Provider Search. It discovers official publisher releases, preserves immutable source provenance, builds complete analytical releases, and serves stable read-only API contracts to downstream products.

One platform combines 16 registered sources from CMS, NPPES, Open Payments, and AACT / ClinicalTrials.gov. Every release is checksum-verified, compared with the prior warehouse, rehearsed in isolation, and promoted as one reversible serving unit.

What it enables

- **Provider discovery:** normalized identity, specialty, taxonomy, enrollment, and location around NPI.
- **Practice intelligence:** group practices, provider rosters, primary sites, and market footprints.
- **Utilization intelligence:** Medicare volume, payments, beneficiary mix, prescribing, DME referrals, and services.
- **Network intelligence:** conservatively inferred hospital relationships with explicit confidence labels.
- **Industry transparency:** reported general, research, and ownership relationships without causal claims.
- **Research intelligence:** investigator evidence and daily AACT / ClinicalTrials.gov snapshots.
- **Market change detection:** new providers, moves, taxonomy changes, reactivations, and deactivations.

Validated production signal	Scale
Registered publisher sources	16
Core provider records	7,373,208
Practice locations	2,341,984
Provider-service records	9,306,818
Provider-drug records	22,444,680
Open Payments general-payment rows	16,131,856
Inferred hospital affiliations	139,775
AACT clinical studies	594,772

Snapshot metrics describe the active validated July 22, 2026 release.

02 Curated data marts

Publisher-shaped raw tables provide auditability. Curated marts provide stable, product-ready semantics while preserving source period and release provenance. NPI is the shared provider identity key.

Data mart	Primary sources	Primary tables	Decision surface
Provider identity and directory	NPPES monthly/weekly; Provider; PECOS	core_providers, raw_nppes	Identity, specialty, taxonomy, public enrollment, and location around NPI.
Practices and rosters	Reassignment; NPPES monthly/weekly	practice_locations	Group membership, provider rosters, primary sites, and geographic footprint.
Medicare utilization	Physician Provider; Part D Provider; DMEPOS	utilization_metrics	Service volume, payments, beneficiary mix, Part D prescribing, and DME referrals.
Services and drugs	Physician Service; Part D Drug	provider_service_detail, provider_drug_detail	Provider-level service mix and medication activity.
Hospital network intelligence	Reassignment; Hospital Enrollments	hospital_affiliations, raw_hospital_enrollments	Conservatively inferred relationships with explicit source and confidence.
Enrollment and eligibility	PECOS; Order and Referring	order_referring_eligibility, PECOS fields	Public Medicare enrollment presence and order/referring eligibility.
Quality and participation	Quality Payment Program Experience	provider_quality_scores	Public QPP participation, practice, and performance measures.
Industry and research	Open Payments general, research, ownership	industry_relationships, kol_summary, Open Payments raw	Reported general, research, and ownership relationships.
New Provider Radar	NPPES monthly/weekly	nppes_radar_provider_state/events/releases	New providers, moves, taxonomy changes, reactivations, and deactivations.
Clinical research	AACT ClinicalTrials.gov snapshot	AACT PostgreSQL ctgov schema	Studies, investigators, sponsors, conditions, interventions, and facilities.

03 Source portfolio and cadence

A schedule is a discovery opportunity, not permission to ingest. A source advances only when primary publisher metadata exposes a different, parseable release and every validation gate passes.

Source	Publisher cadence	Platform policy	Validated period
Physician & Other Practitioners - Provider	Annual	Daily check / 48h	2024
Physician & Other Practitioners - Service	Annual	Daily check / 48h	2024
Part D Prescribers - Provider	Annual	Daily check / 48h	2024
Part D Prescribers - Drug	Annual	Daily check / 48h	2024
DMEPOS - Referring Provider	Annual	Daily check / 48h	2023
Quality Payment Program Experience	Annual	Daily check / 48h	2024
PECOS Public Provider Enrollment	Quarterly	Weekly check / 72h	Q1 2026
Order and Referring	About twice weekly	Daily check / 48h	Jul 12-18, 2026
Hospital Enrollments	Monthly	Daily check / 48h	May 2026
Revalidation Group Reassignment	Monthly	Daily check / 48h	Jul 2026
NPPES Monthly V2	Monthly full	Reconcile each release	Jul 13, 2026
NPPES Weekly Incremental V2	Weekly	Apply in period order	Jul 13-19, 2026
Open Payments General	Annual + correction	Weekly; daily in windows	2025
Open Payments Research	Annual + correction	Weekly; daily in windows	2025
Open Payments Ownership	Annual + correction	Weekly; daily in windows	2025
AACT ClinicalTrials.gov Snapshot	Daily	Stage when available	Jul 21, 2026

04 NPPES: three operating loops

Loop	Operating rule
Weekly change detection	Apply each incremental in publisher-period order; emit idempotent Radar events for new providers, moves, taxonomy changes, deactivation, and reactivation.
Monthly authoritative reconciliation	Replace the baseline from every full V2 snapshot; reconcile all NPIs and validate prior weekly state. Weekly files never substitute for the full refresh.
Daily targeted verification	Use the Registry API only for selected NPIs and confidence labeling. It is not a bulk source and cannot advance the installed release.

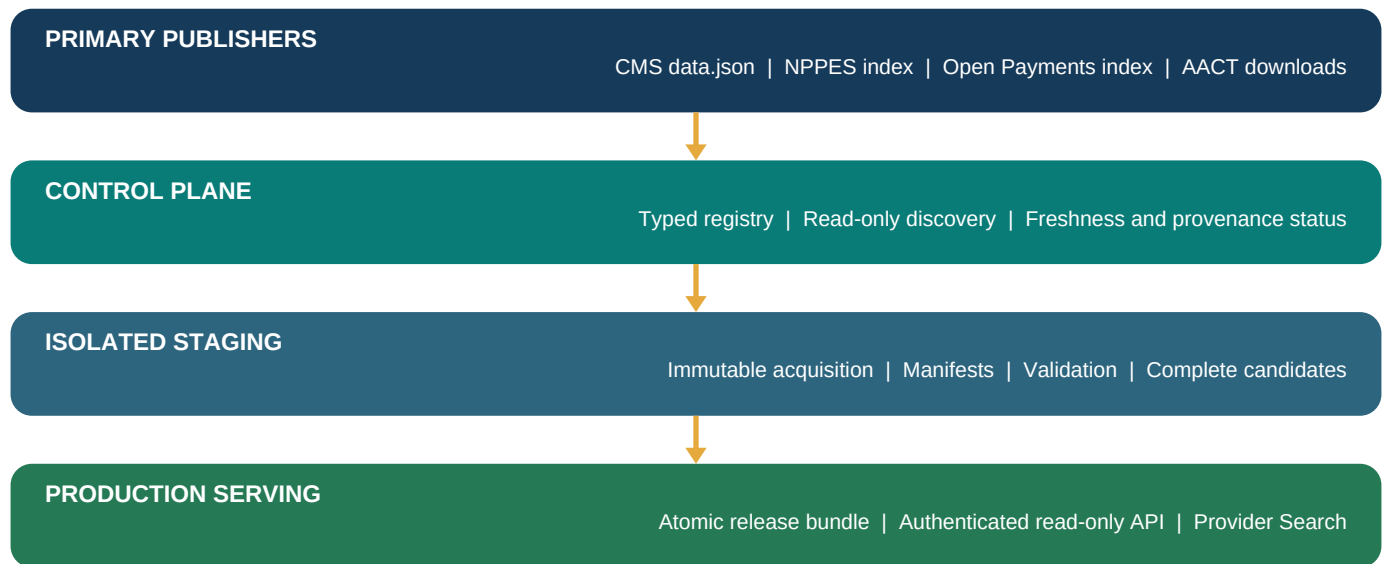
An unexplained gap or overlap blocks the incremental chain until resolved or until the next full monthly snapshot establishes a new baseline.

Primary publisher metadata

- CMS machine-readable catalog: data.cms.gov/data.json
- NPPES downloadable-file index: download.cms.gov/nppes/NPI_Files.html
- Open Payments dataset index: openpaymentsdata.cms.gov/datasets
- AACT download index: aact.ctti-clinicaltrials.org/downloads

05 System architecture

The platform separates read-only discovery, isolated build work, and immutable serving. The API never performs a refresh and the product repository never becomes a bulk-data ingestion engine.

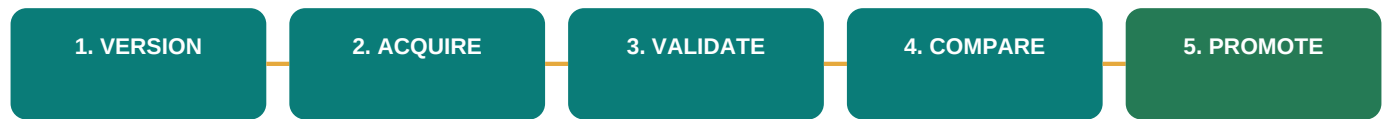


Clear ownership boundaries

Component	Owns	Does not own
cms-data	Discovery, acquisition, validation, releases, rollback, and read-only API	Product UI, workflow state, PHI
provider-search	Product behavior, access plans, UI, workflows, and contract checks	Bulk ingestion or canonical warehouse builds
cms-public-data-catalog	Metadata reference	Runtime ingestion or production data
healthcare-ai	Separate experimental/private-data work	Canonical public-data warehouse

06 End-to-end release process

Every source and every serving release moves through explicit proof gates. Unknown, unavailable, or malformed publisher metadata can never be interpreted as current.

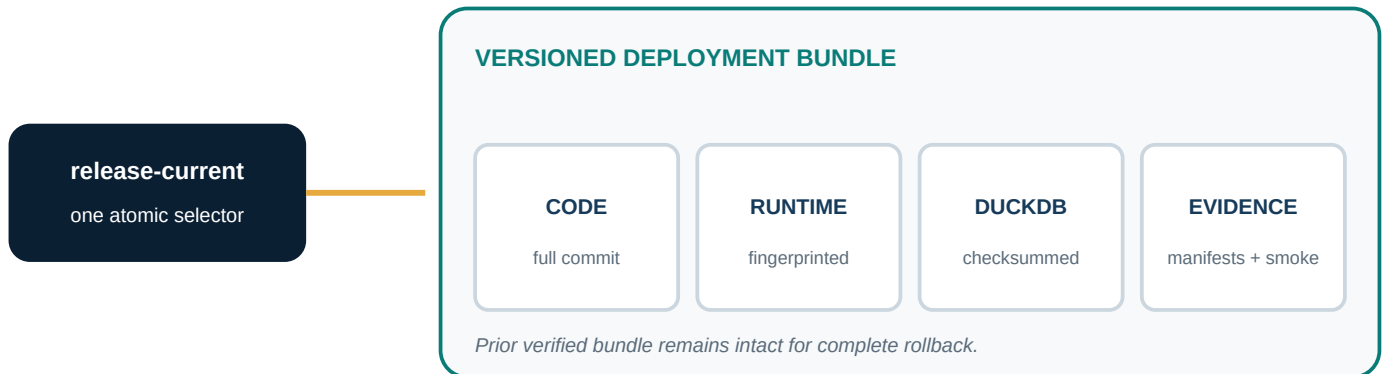


ANY FAILED GATE retains evidence, rejects the candidate, and leaves production unchanged

Stage	Control
1. Discover	Parse primary publisher metadata without downloading bulk files.
2. Acquire	Write a new immutable staging run and enforce transfer, archive, and schema limits.
3. Prove	Record URL, version, source period, timestamps, bytes, SHA-256, schema fingerprint, rows, and code commit.
4. Validate	Check required fields, identifiers, uniqueness, row bounds, period semantics, and source-specific invariants.
5. Build	Construct a complete new DuckDB candidate from a checksum-verified baseline; restore AACT to a versioned PostgreSQL candidate.
6. Compare	Open baseline and candidate read-only; permit only intended table changes and require exact evidence agreement.
7. Rehearse	Run the candidate API on a temporary loopback port and prove API plus process identity.
8. Approve	Require an explicit promotion decision after all artifacts and rollback state are sealed.
9. Select	Atomically replace one release-current bundle pointer; never overwrite a database file.
10. Verify	Restart once, run the full bounded smoke suite, and automatically restore the predecessor on failure.

07 Production release and rollback

Production selects one immutable bundle. That bundle fixes code, Python runtime, DuckDB, and deployment evidence as a coherent unit. The active database is never modified or overwritten in place.



Combined DuckDB and AACT cutover

AACT is PostgreSQL-backed and cannot share a filesystem-atomic transaction with the DuckDB bundle pointer. The safe boundary is API-stopped: a transition sentinel blocks systemd startup, the prior AACT database remains under a versioned rollback name, both selectors are verified before startup, and failure restores both data systems before the API returns.

Failure posture

- Incomplete pointer transitions block startup instead of serving a mixed release.
- Smoke evidence is deployment-specific and time-bounded; stale evidence cannot verify a new release.
- Rollback restores the complete predecessor: code, runtime, DuckDB, AACT snapshot, marker, and source evidence.
- The untouched predecessor remains available after a successful promotion.

08 Read-only serving contracts

Contract family	Representative capabilities
Provider profiles and search	Identity, specialty, taxonomy, public enrollment, utilization, and enriched profiles
Practice intelligence	Specialty capabilities, practice search, rosters, site profiles, and market snapshots
Industry relationships	Open Payments search, normalized options, company summaries, and provider detail
Research evidence	Investigator matching and research-payment evidence
Clinical trials	Exact AACT version identity and study search
New Provider Radar	Market-change events with source release and effective date
Explorer and matching	Curated catalog, bounded samples, unified search, and entity matching

Production smoke contract

Every promotion proves process code, runtime, and database identity; health and authentication; Provider Search practice and profile contracts; Open Payments; research and clinical trials; required tables; and exact expected row counts.

Public data, private boundaries

The warehouse is a public-data plane. Private customer claims, PHI, uploaded client datasets, and product workflow state require separate storage, access controls, retention policies, and contractual review.

09 Trust, interpretation, and attribution

- **Primary publishers only.** Releases come from official CMS, NPPES, Open Payments, and AACT metadata. Dated archive URLs are never guessed.
- **Unknown stays unknown.** File modification time is not provenance, and missing evidence cannot be promoted to current by inference.
- **Immutable lineage.** Source identity, version, period, timestamps, URL, bytes, hash, schema fingerprint, rows, code commit, validation, and promotion state are recorded.
- **Conservative affiliations.** Hospital relationships are inferred only when normalized hospital name and state identify one hospital NPI; ambiguous keys are excluded.
- **No production writes from the API.** Refreshes happen in staging and serving remains read-only.

Data-use guardrails

Source	Required interpretation
CMS and NPPES	Retain attribution; do not imply government endorsement. An NPI does not validate licensure or credentials.
Open Payments	Describe reported transfers of value; do not imply endorsement, causation, or misconduct.
ClinicalTrials.gov / AACT	Attribute sources, show processing date, disclose modifications, and do not market to registry email addresses.
HCPCS Level I	Commercial exposure remains blocked until an appropriate AMA license or approved filter is confirmed.

Official metadata endpoints

- <https://data.cms.gov/data.json>
- https://download.cms.gov/nppes/NPI_Files.html
- <https://openpaymentsdata.cms.gov/datasets>
- <https://aact.ctti-clinicaltrials.org/downloads>

Architectural and operating overview. Snapshot metrics and source periods reflect the active validated July 22, 2026 production release. Last reviewed July 22, 2026.